

## Transformer Model: 30 Q&A

### 1. Q: What is a token in language models?

A: A token is a unit of text, such as a word or subword, used as input to language models. Tokenization splits text into tokens.

### 2. Q: What is embedding in the context of Transformers?

A: Embedding is a vector representation of tokens that captures their semantic meaning in a continuous space.

### 3. Q: What is self-attention?

A: Self-attention is a mechanism that allows each token in a sequence to attend to every other token, enabling contextual understanding.

### 4. Q: What is the purpose of positional encoding?

A: Positional encoding injects information about the position of tokens in a sequence since the Transformer has no recurrence.

### 5. Q: Why use Transformers over RNNs?

A: Transformers enable parallelization, handle long-range dependencies better, and scale more efficiently than RNNs.

### 6. Q: What is multi-head attention?

A: Multi-head attention allows the model to focus on different positions and aspects of the input simultaneously.

### 7. Q: What is the role of the feed-forward network in Transformer blocks?

A: It processes the output of the attention mechanism and helps capture complex transformations of the data.

### 8. Q: How is the attention score computed?

A: It is computed using the dot product of the query and key vectors, scaled, and passed through a

softmax to get attention weights.

**9. Q: What are Q, K, and V in attention?**

A: Q (Query), K (Key), and V (Value) are linear transformations of input embeddings used to calculate attention scores and outputs.

**10. Q: What is the final output of a Transformer layer?**

A: It is a transformed sequence of embeddings after attention and feed-forward processing, used for prediction or passed to the next layer.

**11. Q: What is the role of the softmax function in attention?**

A: The softmax function normalizes attention scores into probabilities that sum to one, which are then used to weigh value vectors.

**12. Q: Why is the dot product scaled in attention?**

A: Scaling prevents large dot product values which can push softmax into regions with very small gradients, harming learning.

**13. Q: What is an attention head?**

A: An attention head is one set of projections for Q, K, and V. Multi-head attention has several such sets to learn from different subspaces.

**14. Q: What are transformer blocks?**

A: A transformer block is a modular layer consisting of multi-head self-attention and position-wise feed-forward networks.

**15. Q: How does attention help with context understanding?**

A: Attention mechanisms allow each token to weigh the relevance of other tokens, capturing dependencies across the sequence.

**16. Q: Why is parallelism important in Transformers?**

A: Parallelism allows efficient computation by processing all tokens simultaneously, unlike sequential RNNs.

**17. Q: How do Transformers generalize to unseen inputs?**

A: Transformers generalize by learning complex patterns in large corpora through layered representations and attention.

**18. Q: What is layer normalization?**

A: Layer normalization standardizes the inputs to a layer to improve training stability and performance.

**19. Q: Why do Transformers use residual connections?**

A: Residual connections help prevent vanishing gradients and enable deeper models by facilitating gradient flow.

**20. Q: What is the main innovation of the Transformer architecture?**

A: Its use of self-attention instead of recurrence, allowing parallel processing and long-range dependency modeling.

**21. Q: How is self-attention computed using matrices?**

A: Q, K, and V matrices are multiplied to get attention scores, which are softmaxed and used to compute a weighted sum of V.

**22. Q: What is the output dimension of multi-head attention?**

A: The outputs from each head are concatenated and linearly transformed to match the model dimension.

**23. Q: How does positional encoding work?**

A: It adds a vector to each input embedding to give the model information about token positions in the sequence.

**24. Q: Why is positional encoding important?**

A: Because attention mechanisms alone are invariant to token position, positional encoding gives a sense of order.

**25. Q: What does 'attention is all you need' mean?**

A: It suggests that self-attention alone is sufficient for sequence modeling tasks, removing the need for RNNs or CNNs.

**26. Q: How are Transformers used in language translation?**

A: Transformers encode the source sentence and decode it into the target language using self-attention and cross-attention.

**27. Q: What is the encoder-decoder structure in Transformers?**

A: The encoder processes the input sequence, and the decoder generates the output sequence using encoder outputs.

**28. Q: How is masking used in Transformers?**

A: Masking prevents attending to future tokens in the decoder and ignores padding tokens in the encoder.

**29. Q: What is the difference between encoder and decoder self-attention?**

A: Encoder self-attention is bidirectional, while decoder self-attention is masked to be causal (unidirectional).

**30. Q: What is the transformer's role in generative AI?**

A: Transformers generate coherent sequences by predicting tokens autoregressively, forming the backbone of models like GPT.